

Computer Network Topologies

Comprehensive Assignment Report & Visualizations

1. Bus Topology

A bus topology consists of a main backbone cable (the "bus") that connects all devices in a network. Data sent from a node travels along the length of the backbone in both directions, and every device on the network receives the message, though only the intended recipient processes it. Terminators are placed at both ends of the backbone to absorb signals and prevent data reflection and signal bounce.

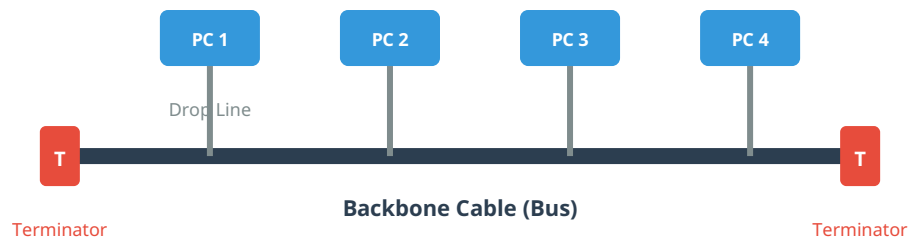


Figure 1: Bus Topology with Central Backbone Cable, Terminators, and Connected Nodes.

Real-World Example: Industrial Automation & Manufacturing Plants (e.g., CAN Bus / Profibus). Bus topologies are frequently used in modern industrial environments and automotive systems where sensors, actuators, and controllers need to communicate along a single twisted-pair wire or coaxial cable running throughout a machinery line or vehicle chassis, minimizing cabling complexity.

2. Star Topology (Cisco Packet Tracer Representation)

In a star topology, all devices (nodes) are individually connected to a central hub or switch. Below is a structural diagram representing the requested Cisco Packet Tracer layout featuring a central switch connected to four PCs named after popular applications: WhatsApp, Zomato, Flipkart, and Spotify.

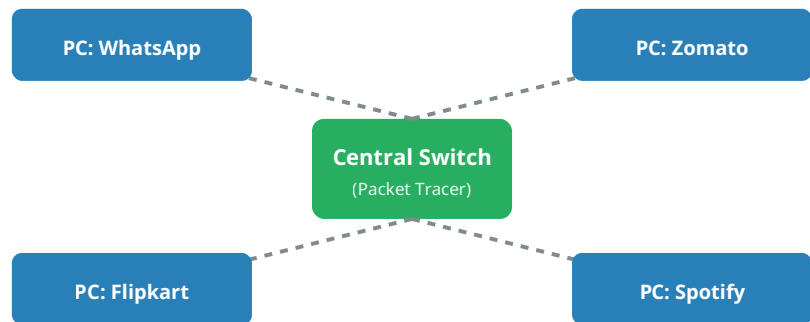


Figure 2: Star Topology Simulation Layout with Central Switch and App-named PCs.

3. Ring Topology & Data Flow

A ring topology connects devices in a circular loop. Data travels from one device to the next around the ring in a single, predetermined direction (unidirectional, either clockwise or counter-clockwise) until it reaches its intended destination device.

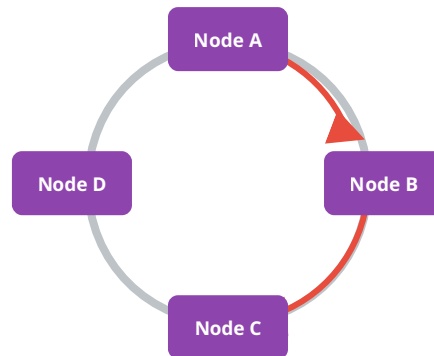


Figure 3: Ring Topology with Unidirectional Data Flow Arrows.

Data Flow Explanation: In a ring topology, data packets travel sequentially from node to node around the closed loop in one direction. Each intermediate device acts as a repeater, regenerating the signal and checking if the packet is addressed to it; if not, it passes the data along to the next neighbor until it reaches the destination.

4. Comparison: Mesh vs. Star Topologies

The table below summarizes two primary advantages and two disadvantages for both Mesh and Star network topologies.

Topology	Advantages	Disadvantages
Mesh Topology	• High Redundancy & Reliability: Multiple pathways exist; failure of one cable or node does not disrupt the network. • High Security & Privacy: Data travels on dedicated point-to-point links, preventing unauthorized interception.	• High Cost & Complexity: Requires significantly more cabling and hardware ports, increasing installation expenses. • Difficult Installation: Configuration and troubleshooting are complex due to the dense interconnectivity.
Star Topology	• Easy Troubleshooting & Isolation: If one cable or node fails, it affects only that single device; the rest of the network operates normally. • Scalability & Ease of Management: Easy to add or remove devices from the central switch without network downtime.	• Single Point of Failure: If the central hub or switch fails, the entire network goes down. • Hardware Dependency: Relies heavily on the performance and capacity of the central device (switch/router).